

The Local Surface Energy Balance and Subsurface Temperature Regime in Antarctica

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ABSTRACT

The physical processes occurring within a perennial snow cover and at its surface are discussed with particular emphasis on monthly variations in Antarctic snow. Results of a one-year simulation of physical processes in snow are compared with 1958 data at Mirny and Pionerskaya, Antarctica. The simulation demonstrates the following: first, solar radiation is effective to a depth of 50 cm or so, but longwave radiation cools the surface alone, causing the "greenhouse" effect first observed in snow at least four decades ago. Highest summertime temperature and initial snowmelt thus occur not at the surface but roughly 10 cm below it. Second, the percolation of meltwater with eventual refreezing during the subsequent winter is the most efficient mechanism for modifying the temperature profile at depths of several meters or more. Third, concerning the surface energy budget, at temperatures below -20°C , atmospheric sensible heat flux approximately compensates net radiation throughout the year. When the temperature nears 0°C (only in summer), the surplus of net radiation is used primarily for changes of phase—sublimation and possibly melting; little energy is used to heat the snow. Then latent heat flux approximately compensates net radiation.

1. Introduction

The vast Antarctic ice sheet, covered by a thin veneer of snow, generates its own climate. Except for man-made objects, mountain ranges, and isolated nunataks, the snow surface is unbroken and unsullied. The albedo is consequently very high, usually above 0.80, making Antarctica the world's coldest continent, even though the daily totals of incident solar radiation in summer are the greatest observed anywhere on earth.

This paper reports on a study of the thermal processes which govern the surface energy balance and the seasonal temperature changes within the snow. We shall refer often to observations made at Pionerskaya ($69^{\circ}44'\text{S}$, $95^{\circ}30'\text{E}$, 2740 m MSL) and Mirny ($66^{\circ}33'\text{S}$, $93^{\circ}01'\text{E}$, 30 m MSL) during the Soviet Antarctic Expeditions, 1956–60 (see Fig. 1). Rusin (1961) has compiled these data and published them in a comprehensive form well suited for the present study.

Observations show clear differences in climate between Mirny and Pionerskaya. Situated on the coast of east Antarctica, Mirny has comparatively mild summers with substantial snowmelt in most years. At Pionerskaya, 400 km inland and high on the glacial slope, temperatures are well below freezing throughout the year; snowmelt has never been observed. Many comparisons can be made between these two stations as

an aid to understanding physical processes in snow, particularly snowmelt.

An essential feature of the present study is a mathematical model which simulates energy transport in snow. In formulating the model, we were guided by the following considerations:

In general, the thermal condition of snow changes in response to energy input at the surface. Solar radiation is the main source of energy during the summer. Another source of energy, effective all year, is downward infrared

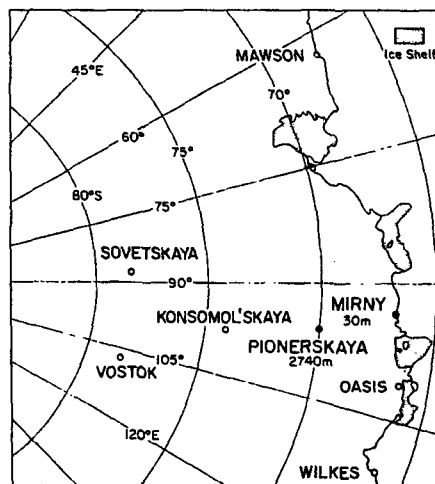


FIG. 1. East Antarctica.

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radiation from atmospheric carbon dioxide, water vapor and clouds. The total annual radiation balance is nonetheless negative over Antarctic snow because solar radiation is virtually nil during the long winter and absorption is but a small fraction of insolation in summer. The annual deficit of the radiative energy budget must be balanced by transfer of sensible and latent heat to the surface from the atmospheric boundary layer.

Snow differs from other solid surfaces in its ability to transmit solar radiation to depths ≥ 50 cm. Cooling due to infrared radiation, however, is confined to a thin surface layer. As a result, a temperature maximum frequently appears 10–15 cm below the surface in summer. This is the “greenhouse” effect observed at least as early as 1932 (Niederdorfer, 1933), theoretically explained by Reuter (1948), and subsequently discussed by Liljequist (1956), Takahashi (1960), Mellor (1964), Geiger (1965), Kondratyev (1969), and others. At Mirny, the greenhouse effect causes subsurface melting even before surface and air temperatures reach 0°C.

Production, percolation and refreezing of meltwater are important thermal processes at Mirny. Energy added to the snow is first stored as sensible heat. Eventually, the heat is sufficient for change of phase. The meltwater thus generated saturates the snow and percolates downward transporting the heat of fusion and markedly raising temperatures at depths where the effects of conduction are hardly noticeable. The snow at Mirny is permeable to 40 m and runoff is negligible so that the melt remains in the snow to refreeze in autumn.

It is clear from the foregoing discussion that an adequate quantitative description of the temperature regime in snow must include penetration and absorption of solar radiation, conduction of heat and moisture, and snowmelt. These processes are expressed mathematically in Section 2. In Section 3 the external heating is formulated and the snow parameters needed for computation are discussed. The numerical technique for integrating the relevant differential equations is described in Section 4. Finally in Section 5, seasonal changes in snow temperature and snowmelt are discussed by comparing computed results with observations.

Two processes not treated explicitly in this paper are convection and snow metamorphosis. Convection is used in the broadest sense to imply bulk motion of air in snow. Measurements by Dubrovin (1961) demonstrate its existence in Antarctic snow. Weller and Schwerdtfeger (1970) argue that convection explains the significant departure of the effective thermal conductivity from the true thermal conductivity below 2 m at Plateau Station, Antarctica, especially in winter when unstable temperature gradients exist in the top few meters. In a laboratory experiment, Yen (1969) showed that effective thermal conductivity could be increased 10–60% depending upon the imposed mass flux of air in the snow sample. Under natural conditions,

however, it is doubtful that convection is of crucial importance in determining the temperature profile.

Snow metamorphosis refers to the continual modification of crystals from the time of deposition to the time of incorporation in glacier ice. It proceeds in a variety of ways—migration of water vapor under the influence of surface free energy or a temperature gradient, cycles of melting and refreezing, compaction from the weight of overlying snow, etc.—but the result is invariably a change in snow density. Strictly speaking, the density stratification is time-dependent. However, since there is no “equation of state” for snow, any connection between snow density and environmental parameters is ignored.

In this paper, we consider only vertical fluxes, because temperature gradients are at least two orders of magnitude smaller in the horizontal than in the vertical.

2. Heat transport in the snow

a. Solar radiation

The two-stream approximation, first proposed by Schuster (1905) and later applied to solar radiation in snow by Dunkle and Gier (1953), is applicable here. Consider an infinitesimally thin snow layer of thickness dz . In passing through this layer, the incident downward flux Y is diminished by absorption and back reflection and augmented by back reflection of the upward flux X . If the vertical coordinate z is positive downward, the total change in the downward flux may be written

$$dY = -aYdz - rYdz + rXdz. \quad (1a)$$

Note that dY depends upon the downward and upward incident fluxes and is proportional to the thickness of the layer. The coefficients a and r refer to absorption and reflection per unit length; in general, they depend upon snow properties. The change in the upward flux is given by

$$dX = aXdz + rXdz - rYdz. \quad (1b)$$

The boundary conditions on (1a) and (1b) are, respectively, $Y(0) = R_i$ and $X(z') = 0$, where R_i is the incident solar flux and z' is a depth below which solar radiation is negligible.

In homogeneous snow, a and r are constant, and Eqs. (1) have the analytic solutions

$$Y = R_i e^{-\mu z}, \quad X = m R_i e^{-\mu z}, \quad (2)$$

where

$$m = \frac{a + r - \mu}{r}, \quad \mu = (a^2 + 2ar)^{1/2}, \quad (3)$$

and μ is the total extinction coefficient. The ratio of upward to downward flux, $X/Y = m$, is constant with depth and is also the surface albedo.

The alteration of the optical properties of snow by melt-freeze cycles is both rapid and pronounced. It is well known that wet snow absorbs more solar radiation

TABLE 1. Climatological data for Mirny and Pionerskaya (monthly means).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
<i>Mirny</i>												
Temperature (°C)	-1.9	-5.2	-9.2	-11.8	-13.2	-16.5	-16.4	-17.9	-18.0	-12.8	-7.6	-2.0
1956-59												
Humidity (% over ice)	67	66	67	68	71	69	74	72	69	67	66	72
1956-59												
Wind speed (m sec ⁻¹)	7.8	9.4	11.5	13.9	14.3	12.5	14.1	14.1	13.5	10.8	9.8	8.2
Feb 56-Mar 60												
Fractional cloudiness	0.66	0.61	0.66	0.66	0.66	0.51	0.70	0.65	0.61	0.68	0.51	0.74
1956-58												
Precipitation (mm melted)	4	9	32	33	105	66	106	78	103	49	13	28
1956-59												
Snow albedo	0.80	0.84	0.87	0.91	—	—	—	0.93	0.90	0.85	0.83	0.81
Apr 56-Dec 59												
<i>Pionerskaya</i>												
Temperature (°C)	-23.4	-32.4	-38.4	-39.2	-42.9	-46.0	-47.3	-48.3	-44.4	-39.3	-31.1	-23.1
1956-58												
Humidity (% over ice)	76	72	71	74	76	76	74	74	76	78	79	80
1956-58												
Wind speed	10.6	11.0	11.0	12.5	9.6	9.6	10.5	11.0	11.5	10.4	9.9	9.4
May 56-Dec 58												
Fractional cloudiness	0.65	0.52	0.62	0.68	0.60	0.50	0.62	0.55	0.60	0.63	0.54	0.66
1956-58												
Precipitation (mm melted)	45	29	64	195	67	85	103	114	117	61	36	49
1957-58												
Snow albedo	0.82	0.83	0.84	0.86	—	—	—	—	0.97	0.87	0.86	0.84
1956-58, record not continuous												

than dry snow; once snow starts to melt, the subsequent change in absorptivity accelerates melting. It is desirable to incorporate this coupling in the present calculations. Toward this end, we assume that both the absorption and scattering coefficients are functions of the liquid water content of snow. Since liquid water content varies with depth, especially in the melting layers, the absorption and scattering coefficients are ultimately functions of z . The solution of Eqs. (1) for this nonhomogeneous case is obtained numerically with the surface insolation as an upper boundary condition (see the Appendix). The net radiative flux is calculated from

$$Q_s = Y - X. \quad (4)$$

The explicit dependence of a and r on liquid water content is discussed in Section 3.

b. Conduction of sensible and latent heat

The sensible heat flux is proportional to the temperature gradient through

$$Q_c = -\lambda \frac{\partial T}{\partial z}. \quad (5)$$

The thermal conductivity λ is, in general, a function of depth. Since the thermal conductivity for air is much less than that for ice, conduction of sensible heat occurs principally within snow grains themselves or between grains in contact. The conductivity, therefore, depends upon grain packing and bonding, i.e., density. Direct measurements of λ at various depths are seldom available in Antarctica, so it is necessary to use an empirical formula relating λ and z through snow density.

A temperature gradient in snow implies a vapor pressure gradient since the air is saturated with respect to a plane surface of ice.³ If the snow temperature is

³ The effect of curvature on saturation vapor pressure is negligible for the grain sizes encountered in Antarctic snow.

close to the melting point, the latent heat transported by means of water vapor conduction can amount to 20% of the sensible heat. Whereas the flux of sensible heat occurs principally by conduction through the grains, the flux of water vapor F_v occurs by molecular diffusion through the air passageways. Thus,

$$F_v = -D_e \frac{\partial \rho_v}{\partial z}. \quad (6)$$

The constant of proportionality D_e is water vapor diffusivity; ρ_v is the density of water vapor. Using the perfect gas law for water vapor, one can derive an approximate expression for the latent heat flux:

$$Q_v = L_s F_v = -\frac{D_e L_s}{R_v T} \frac{d e_{si}}{dT} \frac{\partial T}{\partial z}, \quad (7)$$

where L_s is the latent heat of sublimation, R_v the gas constant for water vapor, and e_{si} the saturation vapor pressure over ice.

If the temperature is below the melting point, the flux divergence of heat appears as a temperature change according to

$$\rho_s c_s \frac{\partial T}{\partial t} = -\frac{\partial H}{\partial z}, \quad T < 273\text{K}, \quad (8)$$

where ρ_s is snow density, c_s the specific heat capacity of snow, and H the total heat flux calculated from (4), (5), and (7) by

$$H = Q_c + Q_v + Q_s. \quad (9)$$

c. Phase changes

There are two distinct responses of snow to input of heat. When the temperature is sufficiently low, the response [Eq. (8)] is warming or cooling. However, when the temperature reaches the melting point, heat

energy is used to break the hydrogen bonds that hold the water vapor molecules in the crystal lattice; there is no further increase in temperature.

The local change of liquid water content l (grams liquid water per gram of unmelted snow) is given by

$$\frac{\partial}{\partial t}(\rho_s l) = -\frac{1}{L_f} \frac{\partial H}{\partial z}, \quad T = 273\text{K}, \quad l < l_{\max}, \quad (10)$$

where L_f is latent heat of fusion, and $l_{\max}$ is the water-holding capacity of snow, the amount of hygroscopic and capillary water that can be held in snow before gravitational flow begins. The right-hand side allows for production and refreezing of meltwater.

Should melting occur in snow that is already saturated, the water in excess of $l_{\max}$ percolates to a non-saturated level. Discussion of percolation is deferred until Section 4.

3. External forcing and snow parameters

The equations described in the last section constitute a model for Antarctic snow if the external boundary conditions and various snow parameters are given. The lower boundary condition is constant temperature; the upper boundary condition is given by the surface energy balance equation

$$H(0,t) = R_s + R_L + H_c + H_E, \quad (11)$$

where $H(0,t)$ is the total energy flux through the surface, R_s the absorbed solar flux, R_L the net longwave flux, H_c the turbulent flux of sensible heat, and H_E the turbulent flux of latent heat. Some of the meteorological variables that affect the surface energy balance are given in Table 1.

The expression for the solar flux absorbed in the snow is

$$R_s = (1 - A)R_i. \quad (12)$$

The albedo A increases with solar zenith angle, especially when the sun is low in the sky (see Table 1). Contributing to the extremely high albedo in much of Antarctica are level or gently sloping terrain, frequent light snowfall and blowing snow, and lack of atmospheric pollution. The incident solar flux R_i is calculated as a function of solar zenith angle under the assumption of exponential attenuation by the atmosphere. An empirical correction for cloudiness suggested by Ångström (1924) is used.

Assuming that the snow surface emits as a gray body in the infrared, we can express the upward longwave flux $R_L \uparrow$ by the Stefan-Boltzmann law

$$R_L \uparrow = \epsilon \sigma T_s^4, \quad (13)$$

where ϵ is snow emissivity and T_s surface temperature. The effective radiation at the surface for clear skies is

$$(R_L)_{\text{clear}} = \epsilon R_L \downarrow - R_L \uparrow. \quad (14)$$

To calculate the downward atmospheric radiation $R_L \downarrow$, we use Brunt's (1932) formula with empirical coefficients derived for Mirny (Rusin, 1961, p. 292). Gavrilova (1963) has investigated the effect of cloudiness on effective radiation in the Soviet Arctic. Based on hundreds of measurements, she determined

$$R_L = (R_L)_{\text{clear}}(1 - k_2 c^2),$$

where c is fractional cloudiness (all levels) and k_2 is an empirical dimensionless constant. In snow-covered polar regions, where inversions are frequent and clouds relatively low, $k_2 \geq 0.9$. This relation fits measurements by Liljequist (1956, p. 156) of effective radiation vs cloudiness; it can be used to represent monthly mean conditions with fair accuracy. The Ångström ratios $R_L/R_L \uparrow$ derived from the above three equations are close to those reported by Schwerdtfeger (1970):

		Computed by model	Given by Schwerdtfeger (1970, p. 260)
Mirny	Summer	0.13	0.09
	Winter	0.15	0.13
Pionerskaya	Summer	0.18	0.17
	Winter	0.19	0.17

The turbulent fluxes of sensible and latent heat H_c and H_E are estimated from

$$H_c = C_H c_p \rho (\theta - \theta_s) u, \quad (15)$$

$$H_E = C_E L_s \rho (q - q_s) u, \quad (16)$$

where:

- C_H, C_E drag coefficients for transfer of sensible and latent heat, respectively
- c_p specific heat of air at constant pressure
- L_s latent heat of sublimation
- ρ air density
- θ, θ_s potential temperature at screen height and the surface, respectively
- q, q_s specific humidity at screen height and the surface, respectively
- u horizontal wind speed at screen height

Values of θ and q are calculated daily from temperatures and dew points in 1958 synoptic reports (WMO, 1962). The surface temperature is computed from the finite-difference form of (8); q_s is obtained from surface temperature by assuming saturation over ice. The effect of stability on the drag coefficient is accounted for by the method of Deardorff (1968). Under adiabatic conditions, drag coefficients for momentum, heat and moisture are assumed equal, that is, $C_D = C_H = C_E = k^2 / [\ln(z_1/z_0)^2]$, where k is the von Kármán constant, z_1 anemometer height, and z_0 the roughness length. The value $C_D = 0.0015$ for neutral stability used in the model compares with 0.0014 determined for Maudheim, Antarctica (Liljequist, 1956, p. 200, $z_1 = 5$ m), and 0.0015 deter-

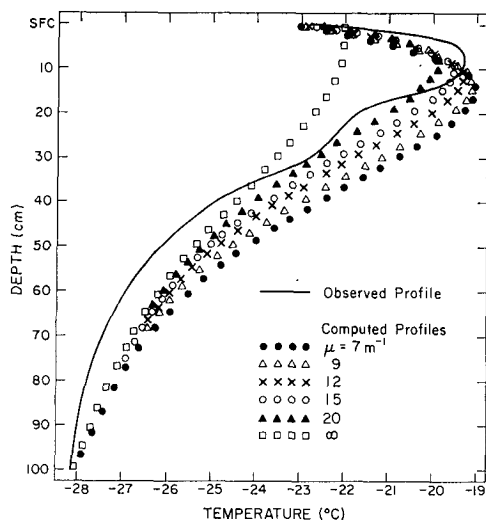


FIG. 2. Mean daily temperatures, computed and observed, for Pionerskaya, December 1956, for various values of the extinction coefficient μ .

mined for the South Pole (Dalrymple *et al.*, 1966, p. 34, $z_1=4$ m). In quoting these values, we assume that $k=0.4$ and that the definition of C_D is the one given above.

We turn now to a discussion of snow parameters.

With the exception of the topmost centimeter, dry Antarctic snow can be considered homogeneous to a depth of one meter, since density and grain size do not vary appreciably in this layer (Stephenson, 1966). It is permissible to use (2) for the net solar flux in snow except during melting, with the ratio of upward to downward flux m determined by the observed mean monthly albedo (Table 1).

The extinction coefficient μ was originally estimated from field measurements at Mawson and Plateau Station, Antarctica, by Weller and Schwerdtfeger (1967, 1970). Their values, ranging from 0.043 to 0.110 cm^{-1} at the surface are among the lowest quoted in the literature (see Mellor, 1965) and are not generally applicable at Mirny or Pionerskaya.

The strength of the greenhouse effect depends greatly on the depth of penetration of solar radiation. At depths of many centimeters, considerable time is needed to dissipate the absorption heat which must first be conducted to the surface where most cooling occurs. Consequently, smaller extinction coefficients are associated with higher temperature in the near-surface snow.

This property has proven useful in determining μ at Pionerskaya. We simulated the diurnal temperature variation in the top meter of snow and compared results for different values of μ with mean 3-hr temperatures at Pionerskaya during December 1956. Resolution in the observed data (Rusin, 1961, p. 38) is sufficient to reveal the greenhouse effect. Computed and observed data

compared favorably when $\mu=0.15$ or 0.20 cm^{-1} (see Fig. 2).

At Mirny, since the diurnal variation of snow temperature is unknown, the value of μ in dry snow had to be determined experimentally. The result was $\mu=0.10 \text{ cm}^{-1}$. Observations made outside Antarctica suggest an inverse variation of μ with grain size (Mellor, 1965) and density (Ambach and Habicht, 1962; Thomas, 1963), although Mellor's data indicate that μ increases with density up to about 0.5 gm cm^{-3} . Nevertheless, experiments for the present study indicate that the high-density, coarse-grained snow at Mirny is more transparent to solar radiation than the snow at Pionerskaya.

During the melt season at Mirny, liquid water content varies with depth, rendering the snow inhomogeneous. In this condition, Eqs. (1) no longer have analytic solutions. Laboratory measurements of a and r as a function of liquid water content are nonexistent, so this dependence is estimated from knowledge of the behavior of m and μ in wet snow. Eqs. (3), valid in homogeneous snow, may be solved for a and r . The solutions are

$$a = \frac{1-m}{1+m} \mu, \quad r = \frac{2m\mu}{1-m^2}. \quad (17)$$

By assuming $\mu=\mu(l)$ and $m=m(l)$, we can determine $a(l)$ and $r(l)$ and thus solve the transfer equations (1) in the nonhomogeneous case; l is known from (10).

Mellor's (1965) data suggest a smaller value of μ for wet snow than for dry snow; certainly, the reflecting capability of a crystal diminishes when it partially melts or becomes coated with water. Our numerical experiments suggest the same: $\mu=0.08 \text{ cm}^{-1}$ gives best results in saturated snow. For nonsaturated wet snow, we use $\mu=0.10-0.02(l/l_{\max})$.

The ratio of upward to downward shortwave flux m is less for wet snow than for dry snow. At Mirny, the value of m for saturated snow is probably near 0.60. In January 1957, unusually large amounts of snow melted at Mirny, and the mean albedo for the month

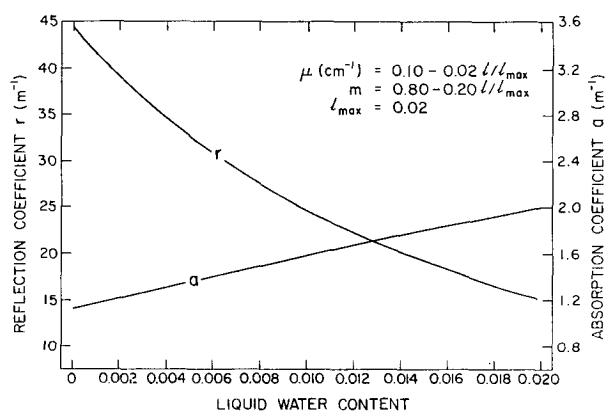


FIG. 3. Values of reflection and absorption coefficients for solar radiation in snow as a function of liquid water content.

was 0.61. The average value of m for dry Antarctic snow is near 0.80. For nonsaturated wet snow, we use $m=0.80-0.20(l/l_{\max})$.

The dependence of the absorption and reflection coefficients on liquid water content (Fig. 3) is reasonable but subject to revision in light of future measurements.

Estimates for the water-holding capacity of snow, $l_{\max}$, vary widely. Gerdel (1954), Kryzhanovskaya (1966), and Glazyrin and Lifanov (1968) have determined $l_{\max}$ in various ways. Their estimates range from 0.007 to 0.300. Tests with the model indicate that the amount of meltwater is little affected by changes in $l_{\max}$ but the depth of penetration varies inversely with $l_{\max}$.

Most investigators have found that infrared emissivity ϵ lies above 0.95 (we use 0.98). A notable exception is Dunkle and Gier (1955) who repeatedly obtained a value of 0.82 for a snow sample from the Sierra Nevada Mountains of California; they feel that the value of ϵ may depend upon grain size and texture. Tests of several values of emissivity between 0.82 and 0.98 indicate that the choice of ϵ in the model is not critical.

Mellor (1964) has listed empirical formulas relating thermal conductivity λ [$\text{cal cm}^{-1} \text{sec}^{-1} (^\circ\text{K})^{-1}$] and snow density ρ_s (gm cm^{-3}); several of these take the form

$$\lambda = c_1 \rho_s^2. \quad (18)$$

Kondrat'eva's formula, with $c_1=0.0085$, is valid for the range of densities encountered in Antarctica. It gives higher values of λ than most other formulas and is used in lieu of an explicit treatment of convection in the snow.

Yen (1965) determined the water vapor diffusivity D_e as $0.65 \text{ cm}^2 \text{sec}^{-1}$, independent of snow density when convection is neglected.

Snow density is specified according to the data of Kopanov (1959) and Kotlyakov (1960). The surface densities at Mirny and Pionerskaya are 0.41 and 0.38

gm cm^{-3} , respectively, and the 5 m densities are 0.16 and 0.49 gm cm^{-3} . The assumed stratification is shown in Fig. 4.

4. Numerical technique

Snow temperatures and liquid water contents are computed using the finite-difference analogs of (8) and (10). Eq. (8) may be rewritten in more explicit form

$$\rho_s c_s \frac{\partial T}{\partial t} = - \frac{\partial}{\partial z} \left[\left(\lambda + \frac{D_e L_s}{R_v T} \frac{de_{si}}{dT} \right) \frac{\partial T}{\partial z} - Q_s \right] \quad (19)$$

as a second-order parabolic partial differential equation in T , mildly nonlinear because of the weak temperature dependence of the coefficients; its finite-difference form is

$$(\rho_s c_s)_k \frac{T_k^{n+1} - T_k^n}{\Delta t} = \frac{1}{\Delta z} \left\{ [\lambda_{k+1} + \gamma_{k+1}^n(T)] \frac{T_{k+1}^n - T_k^n}{\Delta z} - (Q_s)_k - [\lambda_k + \gamma_k^n(T)] \frac{T_k^n - T_{k-1}^n}{\Delta z} + (Q_s)_{k-1} \right\}, \quad (20)$$

where

$$\gamma_k^n(T) = \gamma \left(\frac{T_k^n + T_{k-1}^n}{2} \right), \quad \gamma(T) = \frac{D_e L_s}{R_v} \left(\frac{1}{T} \frac{de_{si}}{dT} \right),$$

$$T_k^n = T[n\Delta t, (k-1)\Delta z],$$

$$k=1, L-1 \quad \text{and} \quad n=1, N.$$

Variables are stored on a staggered grid with the temperature and total heat flux at alternate levels. The boundary conditions are given by T_L^n , the fixed temperature at the bottom (15 m), where the annual temperature cycle is no longer detectable, and by H_1^n , the total heat flux at the surface computed from (9) with T_1^n used as the surface temperature.

Whenever $0 < l \leq l_{\max}$ or (20) predicts $T > 0^\circ\text{C}$, we use (10) on the basis of two assumptions: first, any excess of the computed temperature over 0°C is a direct measure of the energy available for melting snow; second, supposing adequate liquid water, any decrement of the computed temperature below 0°C is a direct measure of the fusion heat liberated in the freezing process.

Whenever the finite-difference analog of (10) predicts $l > l_{\max}$, it is assumed that the excess meltwater percolates to a non-saturated level in time Δt . With $l_{\max}=0.02$ and $\rho_s=0.40 \text{ gm cm}^{-3}$, 1 mm of liquid water will saturate 12.5 cm of snow. Even at the extreme melting rate of 1 mm per time step (24 min), a transmission rate of 0.5 cm sec^{-1} is sufficient to remove excess meltwater about as fast as it is produced in the melting layer. Most transmission rates reported in literature (Gerdel, 1954; Anderson, 1968; Glazyrin and Lifanov, 1968; Colbeck, 1971) are above 0.5 cm sec^{-1} .

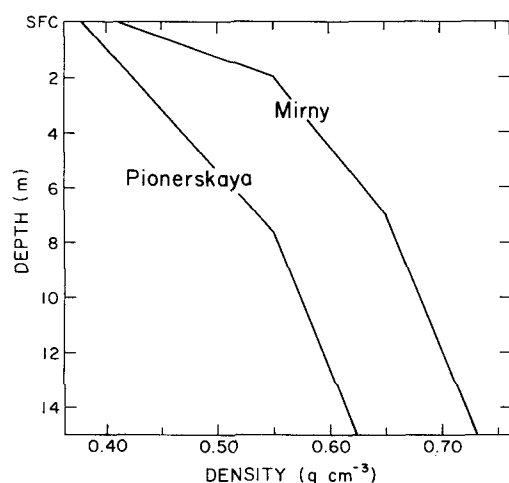


FIG. 4. Density profiles assumed for Mirny and Pionerskaya.

TABLE 2. Origin of data used for comparison with computed results (derived from Rusin, 1961).

Parameter	Period of Measurement		Remarks
	Mirny	Pionerskaya	
Direct solar flux at the surface (S)	3 years (dates not given)	2 years (dates not given)	measured directly (clear sky conditions)
Absorbed solar flux (R_s)	Apr 56-Dec 59	Jul 56-May 57, Apr-Dec 58	calculated as difference between hemispheric downward and reflected solar fluxes, both measured
Longwave balance (R_L)	not applicable	not applicable	calculated from $R_L = R_N - R_s$
Surface longwave emission ($R_L \uparrow$)	Feb 56-Feb 59	May 56-Dec 58	calculated from observed monthly mean temperatures
Total radiation balance (R_N)	Mar 56-Dec 58	Feb-May 57, Apr-Dec 58	measured directly
Sensible heat flux (H_e)	Feb 56-Dec 58	May 56-Dec 58	calculated from measured air temperature, surface temperature, and wind speed
Latent heat flux (H_E)	Feb 56-Dec 58	May 56-Dec 58	calculated from measured air humidity, surface temperature, and wind speed
Total heat flux through snow surface, (H)	not applicable	not applicable	calculated from $H = R_N + H_e + H_E$

The algorithm which keeps track of the liquid water content in snow (Fig. 5) is original. It is based on (10) and the foregoing discussion. There is little reason for using more sophisticated equations since detailed information on the physical state of the snow is lacking.

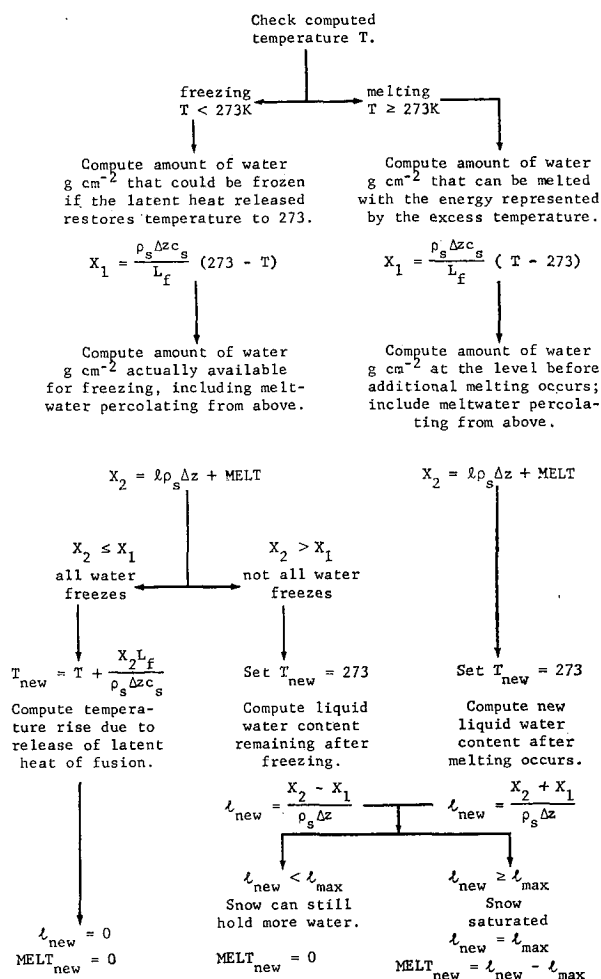


FIG. 5. Algorithm for computing phase changes and percolation of meltwater in the snow.

Numerical simulation begins with data for mid-winter, when temperatures in the entire profile are well below 0°C; thus, computation of snowmelt at Mirny is not prejudiced by initial conditions. The initial temperature profiles at Mirny and Pionerskaya were obtained by averaging the mean monthly temperatures for June and July. Temperatures at Mirny are available only to 6.1 m during 1958 (Rusin, 1961); values at greater depths are estimated from a profile for Mirny by Bentley *et al.* (1964).

The coordinate system extends from the surface to 15 m. We account for snow accumulation and melting by shifting the z axis upward or downward by an appropriate amount.

Eqs. (10) and (20) were integrated for a whole year from 0000 GMT 1 July with $\Delta z = 10$ cm and $\Delta t = 24$ min. Monthly statistics were generated, and these are compared with observed data in Section 5.

5. Seasonal temperature changes and snowmelt

The model was used to simulate conditions in 1958; however, observed data for 1958 alone are not generally available for comparison with computed results. Table 2 gives the periods during which various parameters were measured. Some of the disagreement between computed and observed data can, of course, be traced to deficiencies of the model; but some of the discrepancies may also be caused by comparison of non-contemporaneous data. In addition, the values of sensible and latent heat flux presented by Rusin (1961) were not calculated directly from observations (see his p. 319); consequently, only the month-to-month changes in these fluxes are strictly comparable with those computed by the model.

Fig. 6 shows that the total radiation balance is a small difference between large numbers. The direct solar flux at the surface with clear skies is very great in summer, exceeding one-fourth the solar constant at Pionerskaya in December. Due to the high surface albedo, little of this energy is absorbed at the surface. At Mirny, maximum absorption occurs in January after the summer solstice. There are two reasons for this: first, there

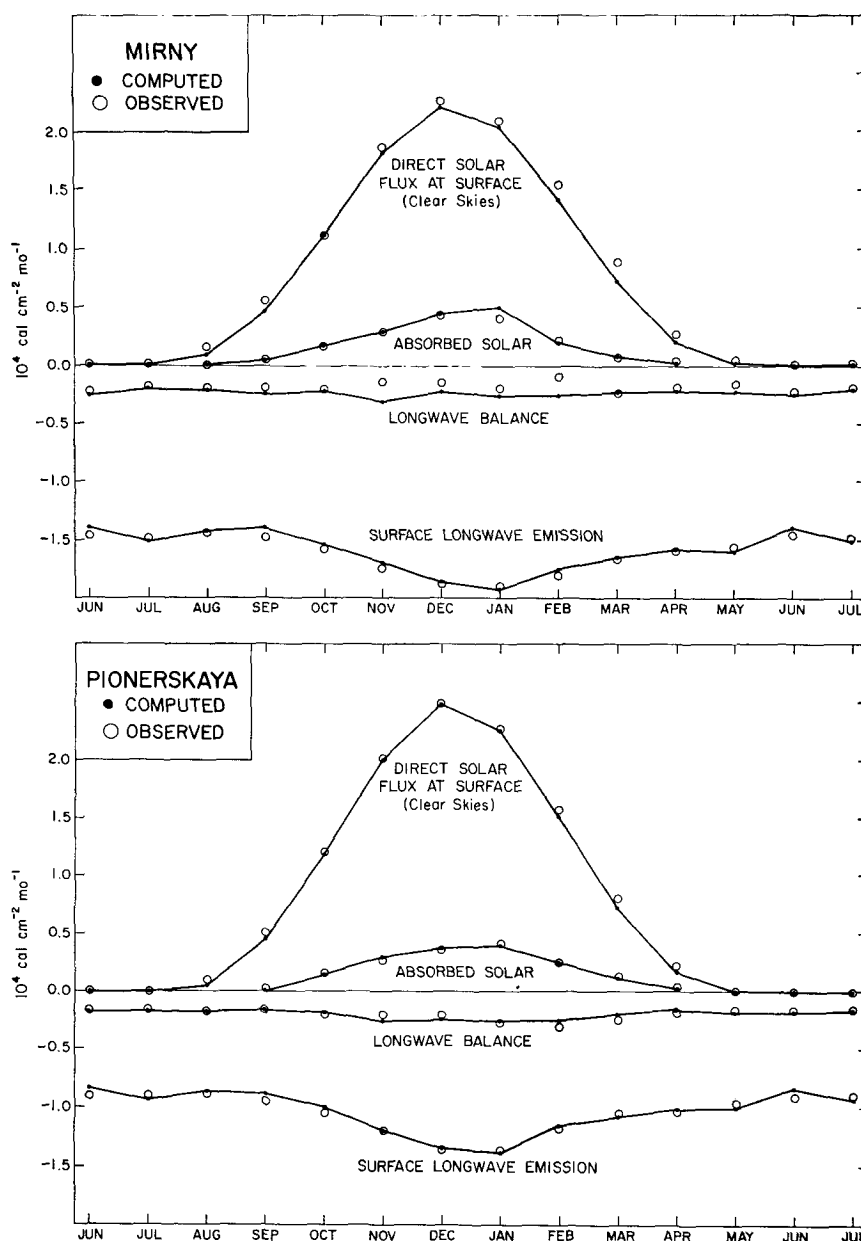


FIG. 6. Components of the radiation balance at Mirny and Pionerskaya, Antarctica. Curves are drawn through computed values (dots). Open circles represent observed values.

are fewer clouds in January than in December and, second, more than two-thirds of the seasonal snowmelt occurs in January, causing relatively low albedo.

Surface longwave emission is 40% greater at Mirny than at Pionerskaya, reflecting a difference of 27C in the computed mean annual surface temperature. Nevertheless, net longwave radiation at both stations is comparable. According to Rusin (1961), Mirny averages 149 overcast days per year; Pionerskaya, 126. Of these, 49 days have low overcasts at Mirny and only 24 at Pionerskaya. The downward atmospheric radiation at Mirny is correspondingly greater.

The monthly variation of the surface heat balance at Mirny is depicted in Fig. 7; note that $H = R_N + H_c + H_E$. The radiation balance R_N has a pronounced seasonal variation but is negative for the year as a whole (net heat radiated to space). The summer maximum of R_N is caused by almost continuous absorption of sunlight in the snow. Considerable melting at Mirny reduces the albedo and enhances absorption. The computed summertime snowmelt (308 mm liquid water) occurs almost exclusively in December (101 mm) and January (206 mm) and compares with an estimated actual summertime snowmelt of 200 mm. The computed sharp peak

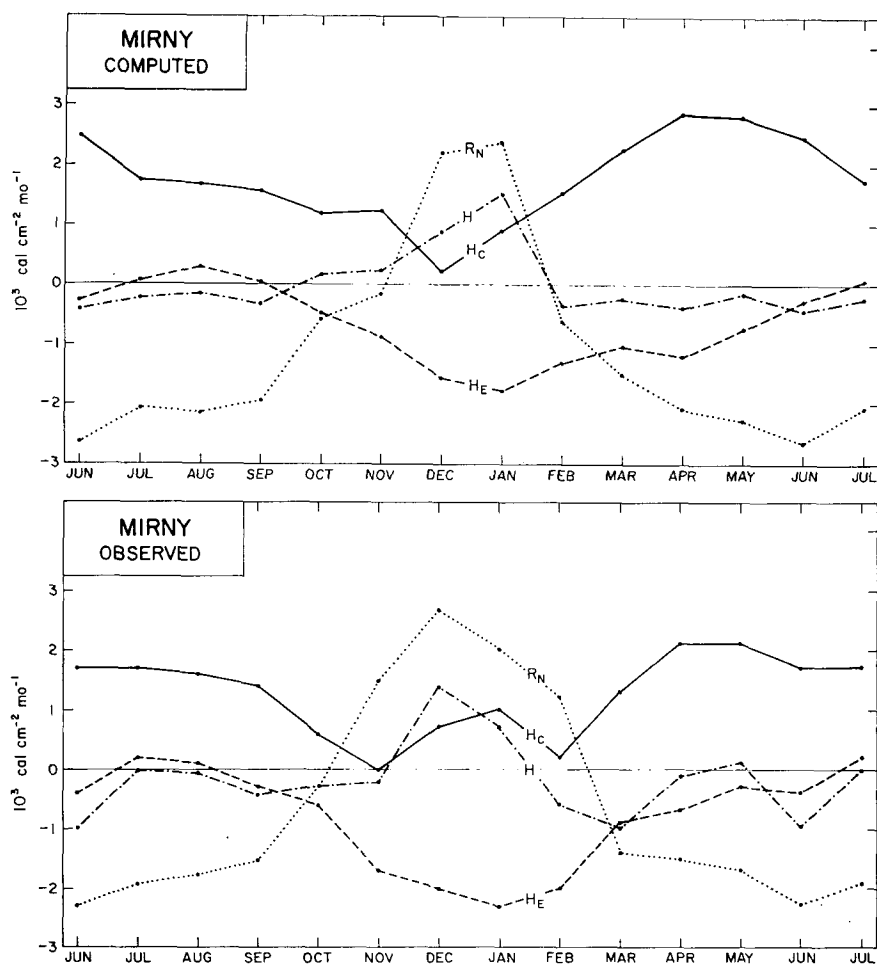


FIG. 7. Surface energy balance at Mirny, Antarctica: R_N , balance of total radiation; H_e , sensible heat flux; H_B , latent heat flux; H , total surface heat flux.

in R_N during these months is responsible for net heat flux H into the snow, and is in agreement with observations.

Atmospheric sensible heat flux H_e is downward for every month of the year at Mirny. During the winter, H_e compensates for net radiative energy loss from the surface. During the summer, at least in more temperate climates, an upward flux of sensible heat would be expected. This does not occur at Mirny because, as the surface temperature nears 0°C , there is a rapid rise in surface vapor pressure. Thus, much of the absorbed solar radiation is used for sublimation. A lesser amount supplies heat of fusion for melting snow. Little solar radiation heats the surface snow layers.

The November and February minima in observed H_e are difficult to explain. Two minima are possible if the air temperature is consistently above freezing during December and January with the surface temperature constrained at 0°C , but measured temperature differences between surface and air (Rusin, 1961, Tables 14 and 206) do not support this.

Atmospheric latent heat flux H_B forms a considerable part of the surface heat balance in summer at Mirny, but plays a minor role in winter because of the low surface vapor pressure associated with low temperatures. For the year as a whole, the computed net evaporative loss is $14\text{--}15\text{ gm cm}^{-2}$; the observed loss is about 18 gm cm^{-2} . These values compare with average annual precipitation of 60 gm cm^{-2} and average summertime melt of 20 gm cm^{-2} .

The surface energy budget at Pionerskaya is given in Fig. 8. Variation of the radiation balance there is similar to the variation at Mirny. Higher summertime albedo (no snowmelt) counteracts greater insolation at Pionerskaya, and the peak in R_N is consequently less impressive than at Mirny. Temperatures as low as -60°C and a well developed surface inversion reduce longwave losses to space during the polar night.

Atmospheric sensible and latent heat fluxes differ between Mirny and Pionerskaya. So little water vapor can be held in the cold air at Pionerskaya that the contribution of latent heat flux to the surface energy budget

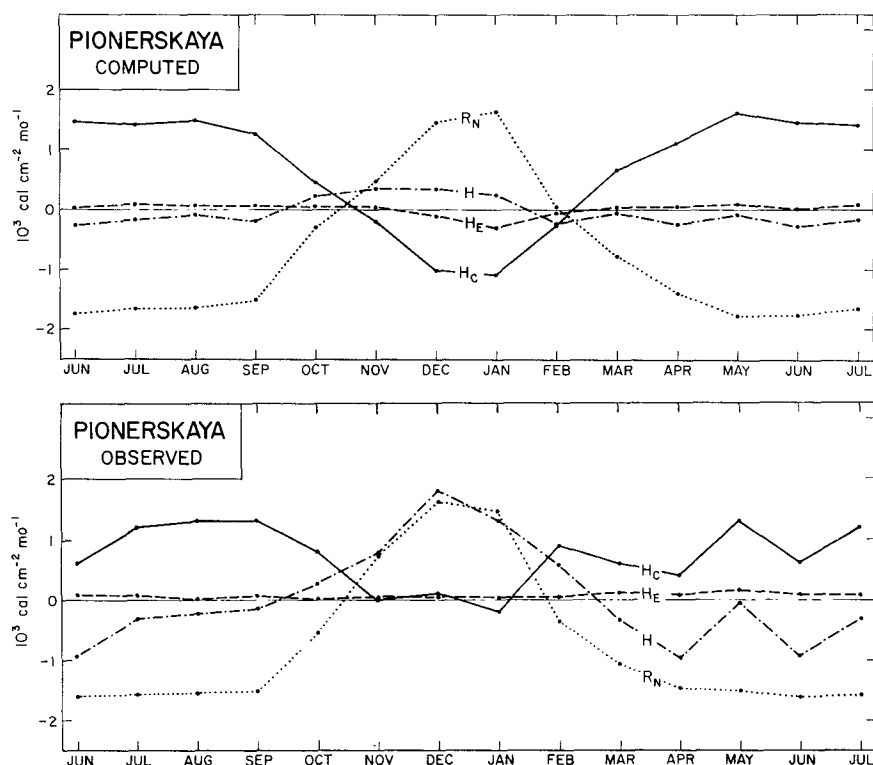


FIG. 8. As in Fig. 7 except for Pionerskaya.

is insignificant. According to dew point measurements [Schwerdtfeger (1970) questions their reliability], hoarfrost should accumulate on the surface snow during every month of the year at Pionerskaya (yearly total less than 1 gm cm^{-2}). The model, however, computes small evaporative losses during the summer months, December–February, which nearly balance the gains from deposition during the rest of the year.

The atmospheric sensible heat flux at Pionerskaya is negative in summer (directed upward). Even though the absolute maximum air temperature in 1956–58 is only -15°C , absorption of solar radiation warms the surface to a temperature higher than that of the air above. The low-level inversion disappears, and small cumulus clouds are observed on occasion, evidence of heating from below. Unstable temperature gradients appear in the lowest few meters even at the South Pole in summer (Dalrymple *et al.*, 1966, p. 39). Unfortunately, no estimates of the sensible heat flux are given for the South Pole in December and January.

Computed and “observed” values of H_c differ markedly from November–July and it is likely that neither is correct during that period. Both computed and observed values should correlate strongly with the difference between screen and surface temperatures ΔT . The mean monthly difference, observed from May 1956 to December 1958 (Rusin, 1961, Table 14), and the simulated difference, derived from 1958 air temperatures

and a computed surface temperature, are plotted in Fig. 9. The month-to-month trends of ΔT agree fairly well, thus casting some doubt on the validity of the “observed” H_c in Fig. 8. On the other hand, the model apparently overestimates $|\Delta T|$. It can be shown, with Deardorff’s (1968) formulation for the drag coefficient, that the error in H_c resulting from a given error in ΔT is larger in the unstable case than in the stable case. For example, if a $|\Delta T|$ of 0.7°K is overestimated by 0.5°K with a wind speed of 5 m sec^{-1} , the error in H_c is about 15% greater in the unstable case.

The disparity between computed and observed values of H in Fig. 8 follows directly from the problem with H_c , since $H = R_N + H_c + H_e$.

It is noteworthy that net radiation and the atmospheric sensible heat flux are comparable in magnitude

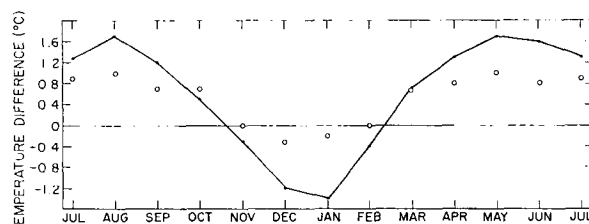
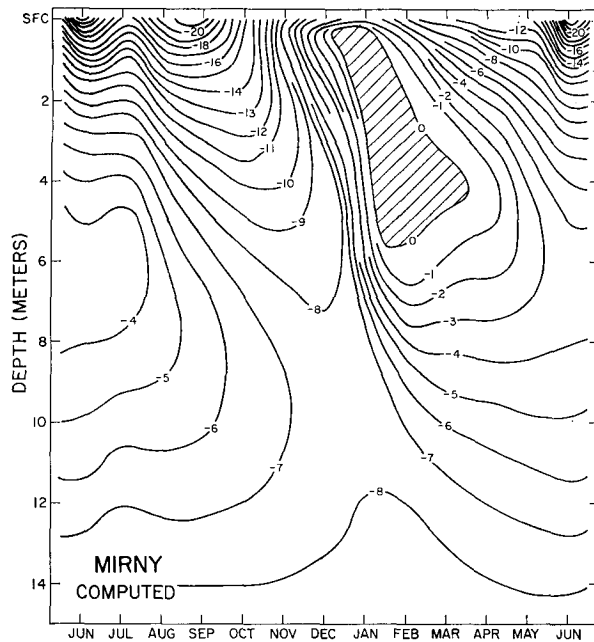
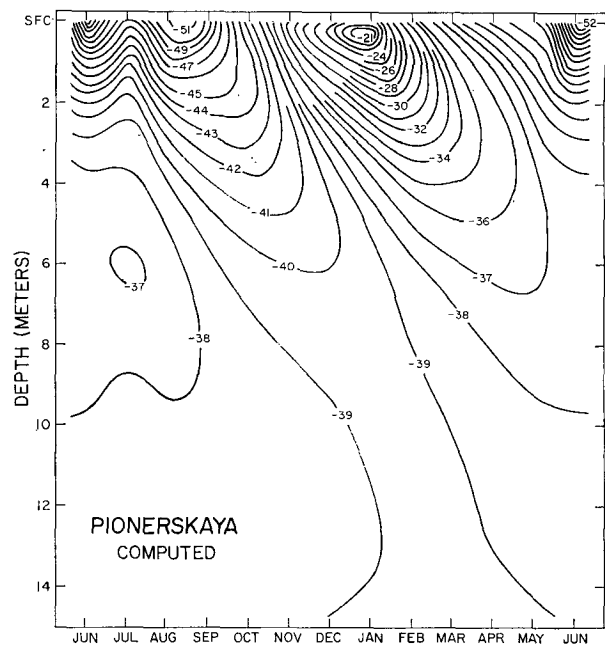


FIG. 9. Computed 1958 (solid line) and observed 1956–58 (circles) monthly mean temperature difference between 2 m and the surface at Pionerskaya.

FIG. 10. Computed snow temperature ($^{\circ}\text{C}$) at Mirny, 1958.FIG. 12. Computed snow temperature ($^{\circ}\text{C}$) at Pionerskaya, 1958.

but opposite in sign for much of the year at Pionerskaya. This is also true of subtropical deserts (see, for example, Sellers, 1965, p. 106, for the monthly surface energy balance at Yuma, Ariz.), but there the similarity ends. Over the Antarctic ice cap, the deficit of net radiation is offset primarily by a loss of heat by the air to the ice; in contrast, the surplus of net radiation over subtropical deserts results in heating of the air by the sand.

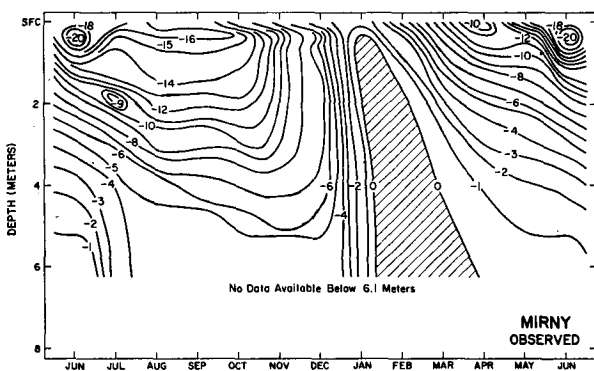
During the winter, the surface energy budgets for Mirny and Pionerskaya are similar, i.e., the net radiative heat loss at the surface is compensated by a downward flux of sensible heat. During the summer, however, the surface energy budget at Mirny is typical of a moist climate (Sellers, 1965, p. 106, West Palm Beach, Fla.), since the surface vapor pressure increases almost exponentially with temperature. The surplus of net

radiation is used primarily for sublimation of snow crystals.

We examine next the computed and observed subsurface temperatures at Mirny (Figs. 10 and 11). It is assumed that the Russians maintained their temperature sensors at constant depth. Most prominent is the isothermal region at 0°C caused by meltwater. Although no data are available below 6.1 m at Mirny, it appears that meltwater penetrates more deeply in the snow than indicated by the model. The sharp alteration of the temperature profile by percolation is obvious; the temperature rise to 0°C propagates from near the surface to a depth of 6 m causing a 10°C temperature change within a single month. The magnitude and rapidity of this change are not even approached in conductive processes in snow.

The main features of the annual temperature variation in the snow are properly simulated by the model, although the computed temperatures are $1\text{--}3^{\circ}\text{C}$ lower than those observed at depths from 4–6 m. If more meltwater had reached this layer in the model, the discrepancy would not have occurred.

Variation of computed and observed subsurface temperatures at Pionerskaya during 1958 is shown in Figs. 12 and 13. The grid distance in the model (10 cm) is sufficiently small to resolve a "greenhouse" effect in December and January when insolation is most intense. No such effect is evident in the observations because temperatures were not measured with sufficient resolution. The same is true for observation of snow temperature at the South Pole (see Dalrymple *et al.*, 1966, Fig. 19). The actual data used in the analysis of Fig. 13

FIG. 11. Observed snow temperature ($^{\circ}\text{C}$) at Mirny, 1958.

appear in horizontal rows. Higher-resolution measurements of the diurnal temperature variation in summer at Pionerskaya clearly show the greenhouse effect (Fig. 14).

Since the annual temperature wave at Pionerskaya is not complicated by snowmelt, it is easy to compare the computed and observed amplitudes and phase lags. Let ΔT be the difference between the highest and lowest monthly mean temperatures. The phase lag in radians (2π is equivalent to one year) and $\ln(\Delta T)$ are plotted on identical scales as a function of depth in Fig. 15. The origins are shifted so that the curves nearly coincide.

The observed amplitude decreases from the surface to 4 m more rapidly than the computed amplitude. Then, from 4–6 m, it decreases more slowly. This curious behavior is most probably due to convection in the snow. Weller and Schwerdtfeger (1970) give evidence that convective processes in Antarctic snow are active between 2 and 6 m. The “hook” at the top of the amplitude profile is the only noticeable effect of solar radiation.

The surface phase lag is considered zero in December and July. At some depths, two circles are plotted, one representing the maximum observed temperature and the other the minimum. Where only one circle appears, the phase lag for the two extremes is the same.

As a check on the internal consistency of the model, we integrate (8) with respect to time from the initial to the final state (from t_i to t_f) and then with respect to depth from the surface to the bottom of the profile. The

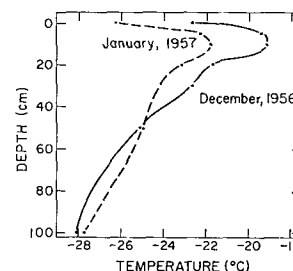


FIG. 14. Greenhouse effect illustrated in mean snow temperatures at Pionerskaya, Antarctica, during December 1956 and January 1957.

result is

$$c_s \int_0^{z_{\text{bot}}} \rho_s [T(z, t_f) - T(z, t_i)] dz = \int_{t_i}^{t_f} [H(0, t) - H(z_{\text{bot}}, t)] dt.$$

The above integrals were evaluated with data generated by the model. The right- and left-hand sides are not equal, but the difference between them, ΔE , is small compared with the total energy passing through the surface during the year, i.e.,

$$\Delta E / \int_{t_i}^{t_f} |H(0, t)| dt < 0.015$$

at both Mirny and Pionerskaya.

6. Implications and generalization of results

The surface balance of total radiation at Mirny and Pionerskaya is negative for the year as a whole. The same is true of all Antarctica except for isolated regions

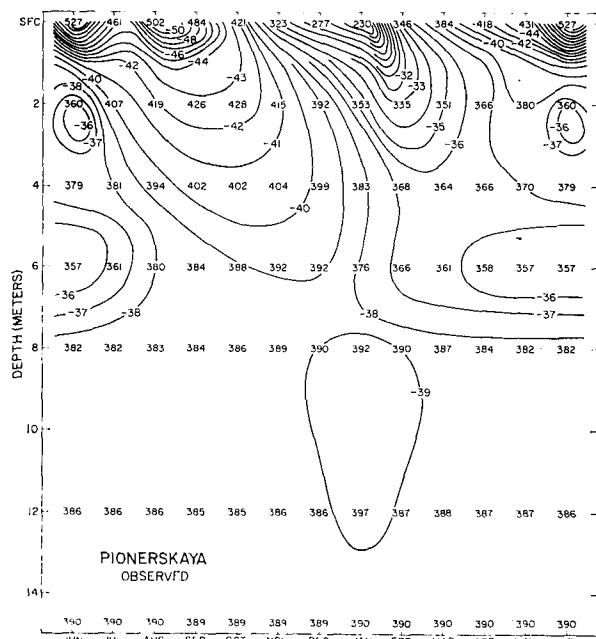


FIG. 13. Observed snow temperature ($^{\circ}\text{C}$) at Pionerskaya, 1958. Observed temperatures for each month appear in horizontal rows in tenths of degrees. Prefix a minus sign.

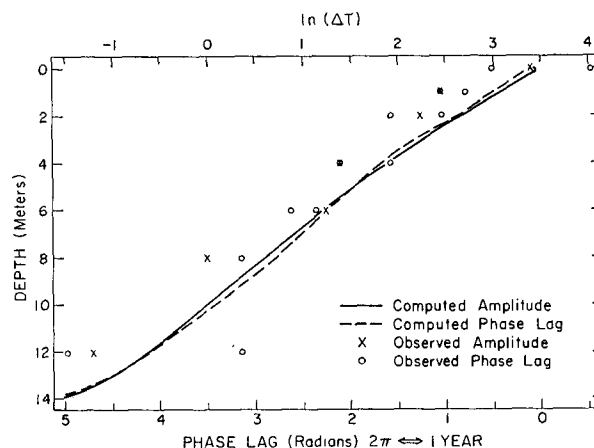


FIG. 15. Computed and observed amplitude and phase lag of snow temperature, Pionerskaya, Antarctica, 1958.

free of ice and snow (Rusin, 1961). The snow surface does not cool indefinitely, however, because of downward transfer of sensible heat from the atmosphere.

The Antarctic continent and the overlying atmosphere act as heat sinks. Heat is supplied from lower latitudes by large-scale transient horizontal eddies (Gabites, 1960; Rubin and Weyant, 1963). There are minor contributions from oceanic currents and advection of latent heat. Rubin (1960) studied atmospheric mass transport across the Antarctic coastline and found inflow in the upper troposphere and outflow near the surface, with mean meridional circulation most pronounced in winter.

As a result of these mass fluxes, downward vertical motion predominates over the Antarctic continent. Adiabatic heating is comparable in magnitude to the poleward transport of sensible heat and helps to maintain the temperature inversion near the surface. It is estimated that one-tenth of the heat lost by the Antarctic atmosphere is lost to the surface through turbulent heat exchange; the other nine-tenths is radiated to space (Rubin and Weyant, 1963).

The model successfully generates the major features of the surface energy balance and the temperature regime in the snow at Mirny and Pionerskaya, but it could be applied at other stations in Antarctica as well. The surface energy budget and snow temperatures have been studied in detail at Maudheim (Liljequist, 1956; Schytt, 1960) and at the South Pole (Dalrymple *et al.*, 1966); extensive data exist for comparison with results.

The model could be applied to snow-covered areas outside Antarctica, but not without modification. Giddings and LaChapelle (1961) have suggested a method for treating solar radiation in shallow snow cover (when the dark earth absorbs heat and causes melting from below). The added complication is minor if vegetation is not a consideration, as over a tundra. The simulation of subsurface temperature is improved, especially at Mirny, by accounting for snowmelt and accumulation. In regions of shallow snow cover, however, this accounting becomes absolutely vital—the surface energy budget depends upon it.

In Antarctica, changes of phase redistribute heat in the snow. In the shallow snow of more temperate climates, meltwater percolates into the soil or runs into streams and rivers. The heat supplied by the sun or the atmosphere to melt the snow does not remain in the snow in this situation, nor is it returned to the atmosphere by evaporation until much later in the hydrologic cycle.

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APPENDIX

Numerical Solution of Equations for Solar Radiation in Snow

Eqs. (1) may be rewritten as

$$Y' = -(a+r)Y + rX, \quad (A1)$$

$$X' = (a+r)X - rY, \quad (A2)$$

with boundary conditions $Y(0) = R_i$ and $X(50 \text{ cm}) = 0$. The prime denotes total differentiation with respect to z , and a and r are functions of z . The solution of (A2) for Y is

$$Y = \frac{1}{r}[(a+r)X - X']. \quad (A3)$$

Substitution of (A3) in (A1) yields

$$Y' = \frac{a+r}{r}X' - \frac{1}{r}(a^2 + 2ar)X. \quad (A4)$$

By differentiating (A2) with respect to z and using (A3) and (A4) in the result, one obtains the following second-order equation in X :

$$X'' - \frac{r'}{r}X' - \left(a' + \frac{r'a}{r} + \mu^2\right)X = 0. \quad (A5)$$

The upper boundary condition obtained from (A2) is

$$X(0) = \frac{1}{a(0) + r(0)}[X'(0) + r(0)R_i].$$

The lower boundary condition is unchanged, namely $X(50) = 0$.

The finite-difference analog of (A5) is

$$\frac{1}{\Delta z} \left(\frac{X_{k+1} - X_k}{\Delta z} - \frac{X_k - X_{k-1}}{\Delta z} \right) - \frac{r_{k+1} - r_{k-1}}{2\Delta z r_k} \frac{X_{k+1} - X_{k-1}}{2\Delta z} - \left[\frac{a_{k+1} - a_{k-1}}{2\Delta z} - \frac{(r_{k+1} - r_{k-1})a_k}{2\Delta z r_k} + \mu_k^2 \right] X_k = 0.$$

which may be rewritten

$$\left(1 + \frac{r_{k+1} - r_{k-1}}{4r_k}\right)X_{k-1} + \left\{ \frac{\Delta z}{2r_k} [a_k(r_{k+1} - r_{k-1}) - r_k(a_{k+1} - a_{k-1})] - [2 + (\Delta z)^2 \mu_k^2] \right\} X_k + \left(1 - \frac{r_{k+1} - r_{k-1}}{4r_k}\right)X_{k+1} = 0, \quad k=2, M-1. \quad (A6)$$

The boundary conditions take the form

$$\text{Top } (k=1): \quad X_1 = \frac{1}{a_1 + r_1} \left(\frac{X_2 - X_1}{\Delta z} + r_1 R_i \right)$$

$$\text{Bottom } (k=M): \quad X_M = 0$$

Eqs. (A6), together with the two boundary conditions, comprise a tri-diagonal system which is easily solved by line inversion (see Wachspress, 1960). Once the X_k , $k=1, M$ are known, the Y_k are found immediately from finite-difference forms of (A3):

$$Y_k = \frac{a_k + r_k}{r_k} X_k - \frac{X_{k+1} - X_{k-1}}{2\Delta z r_k}, \quad k=2, M-1, \quad (A7)$$

$$Y_M = \frac{a_M + r_M}{r_M} X_M - \frac{X_M - X_{M-1}}{\Delta z r_M}. \quad (A8)$$

Note that albedo of snow, $A = X_1/Y_1$, is a by-product of these calculations; the net solar flux in the snow is simply $Q_s = X - Y$.

To test the above scheme, we used the finite-difference equations for a homogeneous case with known analytic solution. The numerical and analytic solutions agreed to within a few percent.

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